

# INTRODUCTION TO HISTORICAL GEOLOGY



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## Licensing

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# 1: What is Historical Geology?

## Introduction

Historical geology seeks to tell the story of the earth. Recall some of your favorite stories. They may have been told through a movie, a show, a book, or by a person.

The planet you live on has some stories to tell. Earth has had 4.55 billion years to rack up experiences that range from ordinary to unimaginably violent. Through a variety of rock-forming processes, Earth has written her autobiography. Historical Geology is the science of reading that autobiography. But to read it, we first need to learn the language in which it is recorded. We must learn to “speak Rock.”

## Stories from Stone

Earth has written countless stories in the language of Rock. This cobble from Montana provides a great example.



Figure 1.1: What is the history of a rock like this? What did it take for this thing to exist? (Callan Bentley photo.)

The texture of this rock contains features we can translate into information. The variety of materials it is made from, how those materials are arranged, other features that cut across them, and the shape of the sample all convey information about past processes and circumstances. What would it take to make a rock just like this one? Why does it exist in this particular way?

As an exercise in practicing decoding Rock, let's translate: The smooth tan material is limestone. It initially formed as inorganic carbonate mud, laid down (1) in calm water, as evidenced by its very fine grain size. It sat still long enough to stick together into a semi-coherent mass. But then it must have been torn up to make (2) mud-chip pieces entombed in (3) quartz sand. The event was energetic enough to rip up the carbonate mud and transport coarse sand particles. Perhaps a hurricane provided the energy. Eventually, the sediment was buried and sank further underground, smothered beneath the crushing weight of thousands of feet of sediment piled atop it and was (4) lithified into rock: No longer loose sediment, it was transformed through compaction and cementation into solid, coherent rock: limestone clasts in quartz sandstone. After it was solid rock, it broke. Some stress, perhaps imparted by the wrenching compression of mountain building, snapped the bonds holding the rock together. This made fractures, but those fractures were soon healed by the precipitation of the mineral quartz, making (5) thin quartz veins. This is something that must have taken place deep underground, but for us to pick it up and hold it, we know it must have been (6) uplifted and brought up to Earth's surface; this particular chunk was broken off from its source outcrop. Finally, it experienced (7) differential weathering, etching away the limestone more readily than the quartz sandstone. This gives the sample its distinctive texture.

The rock is no longer just a rock. By speaking its language, we have turned it into a book, with seven chapters.





Figure 1.2: Points of annotation showing the 7-chapter history of this one rock sample. (Callan Bentley photo.)

This is the power of Historical Geology: it allows curious humans to interact with inert rock, and coax stories from that rock — vital tales of the planet’s ancient past. Being able to decode the stories of rocks, hidden in plain sight, is an ability akin to an acquired superpower. As Spiderman gained his powers after being bitten by a radioactive spider, you are about to be bitten by the Historical Geology bug. What you learn will change the way you look at your world.

Like the best stories from our favorite storytellers, what we learn resonates with us and changes us. The Earth’s many tales help us understand our present, live our lives more meaningfully, and prepare for a future we can better comprehend. The perspective granted by speaking Rock helps us live better lives and look to the future with hope and confidence.

Decoding the handful of past processes encoded into a single cobble is powerful. However, more power still comes from looking at sedimentary rock in outcrop (see figure below), and comparing the conditions of one layer with those below and above it. The common language of historical geology is sedimentary rock, and vast epic tales have been inscribed in the sequence the rock layers. Here is an example, from Inuyama, Japan:



Figure 1.3: Outcrops in Inuyama, Japan, show a Triassic sequence of sedimentary strata that record changes in deep ocean oxygen levels after the end-Permian mass extinction event. The strata have been tectonically rotated due to accretion with the Japanese mainland. (Photo by Yukio Isozaki; reproduced with permission.)

Here, the strata are not in their original horizontal orientation. They have been rotated during tectonic accretion (adding) of these deep ocean strata to the Japanese mainland. The layers get younger to the right.

The important thing is to note that a distinctive color change occurs across the outcrop: the layers shift from black to red. This is a signature of changes in the amount of oxygen in the deep ocean: there was almost none (black), then plenty (red). This is interesting because a major mass extinction occurs just before this sequence at the end of a period of time called the Permian: the end-Permian mass extinction (also called “The Great Dying”). These strata are 247 to 242 Ma, early middle Triassic (i.e. the period after the Permian), showing recovery after the mass extinction. Evidence such as this sequence of strata suggests that ocean anoxia (low oxygen levels) was associated with this major die-off in animals, either as a cause or a consequence. And most everything on the planet died: 95% of species. It was the worst thing to ever happen to life on our planet, and this sequence of black layers transitioning into red layers is trying to tell us why it happened. As we see here, no single layer or rock tells the whole story – we must look at the succession of layers over time.

Why should you care about any of this?

You are an Earthling. You were born on this planet and will live your entire life on its surface. The atoms from which your body is constructed are atoms that have been on this planet for more than four billion years, repurposed endlessly before becoming part of you, and destined for continued eternal recycling after you die. The carbon that makes up your eyeballs, neurons, and fingertips was derived from food you ate, and the plants that captured that carbon pulled it from the atmosphere. It got there perhaps from volcanic emissions, the rotting of a dinosaur, or the combustion of ancient coal. You are literally made of Earth bits. When you eat or breathe or pee, you are connecting with the Earth.



Figure 1.4: We instinctively connect with the Earth from a very young age. (Callan Bentley photo.)

Understanding Historical Geology deepens our appreciation of the planet we interact with throughout our lives. It provides a framework for the existence of all living things, an explanation for how life has evolved over time, and the ability to make predictions for our future Earth.

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## CHAPTER OVERVIEW

### 2: An Introduction to Earth

#### Introduction

The changes that have occurred since the inception of Earth are vast and significant. From the introduction of oxygen into the atmosphere, the progression of life forms, the assembly and deconstruction of several supercontinents, to the extinction of more life forms than exist today. Having a general understanding of these changes can put present change into a more rounded perspective.

How and when does the story of Earth begin? A logical place to start is with the formation of our planet. However, the formation of our planet is part of a larger story, the story of the formation of our solar system.

This chapter summarizes the origin of our solar system and introduces the earliest chapters of the Earth's story.

[2.1: The Origin of Earth and Our Solar System](#)

[2.2: Earth's Structure and the Origin of the Continents and Oceans](#)

[2.3: Key Points](#)

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## 2.1: The Origin of Earth and Our Solar System

The prevailing scientific explanation for the origin of the Earth can be applied to the formation of the Sun and all the other planets. So, it's not only Earth's story but the story of the origin of the whole solar system. Not only that, but our Sun is one star among a hundred million in our galaxy, and our galaxy is one of perhaps a hundred million in the universe. So the lessons we learn by studying our solar system can likely be applied more generally to the formation of other solar systems elsewhere, including those long ago, in galaxies far, far away - the present is the key to the past.

Using powerful telescopes, we observe many other stars, but we also observe other things, including fuzzy-looking "stars" called nebulae. A **nebula** is a massive cloud of gas and dust in space that is the byproduct of a previous star that suffered a supernova explosion - its leftover material. Orion's "belt" is part of a constellation of stars easily identifiable in the northern hemisphere's winter night sky. Below the row of three stars of the belt is a smaller trio of light spots; this is Orion's sword scabbard. Binoculars and a dark enough sky will let you examine these objects for yourself; you will discover that the middle point of light in this smaller trio is not a star. It is a nebula called Messier 42.

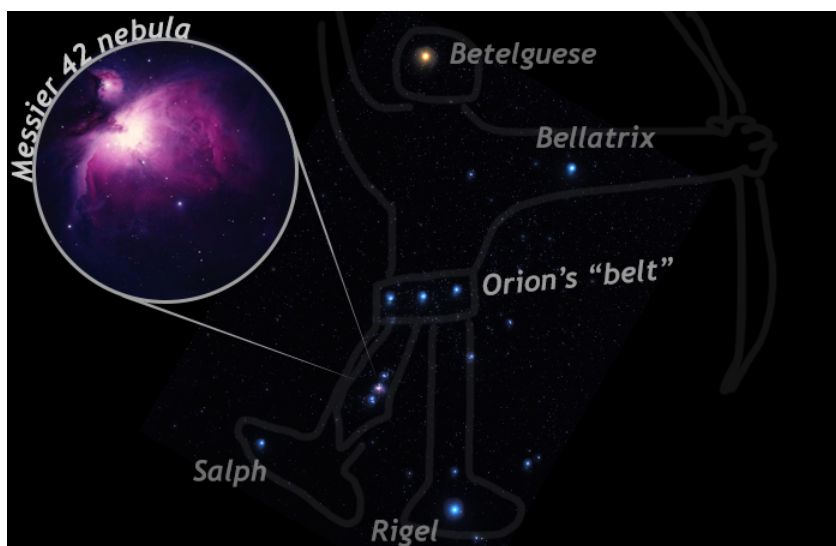


Figure 2.1.1: The Messier 42 nebula, shown in the context of the "scabbard" of the constellation Orion.

Nebulae like Messier 42 are common features of the galaxy, but not as common as stars. Nebulae appear to be short-lived features, as matter is often attracted to other matter. Particles attract one another under the influence of forces such as electrostatic attraction or "static cling". This is the same force that turns bits of dust or fabric into larger "dust bunnies" under your couch!



Figure 2.1.2: Three dust bunnies under the couch: a household hassle, or a glimpse into the birth of the solar system?

Electrostatic forces are effective at attracting small particles over short distances. However, if you want to make big things like planets and stars out of a nebula, gravity needs to take over at some point. Although gravity is a relatively weak force - you overcome the force of gravity every time you pick up your foot to take a step - gravity can work very efficiently over distance if the masses involved are large enough. So static cling initially organized matter into "space dust bunnies" that once large enough, were



pulled together by gravity. The net result is that the countless tiny pieces of the nebula were drawn together, swirling into a denser and denser amalgamation (clump). The nebula began to spin, flattening from top to bottom and forming a rotating disk, like a fried egg with the yolk surrounded by the white.

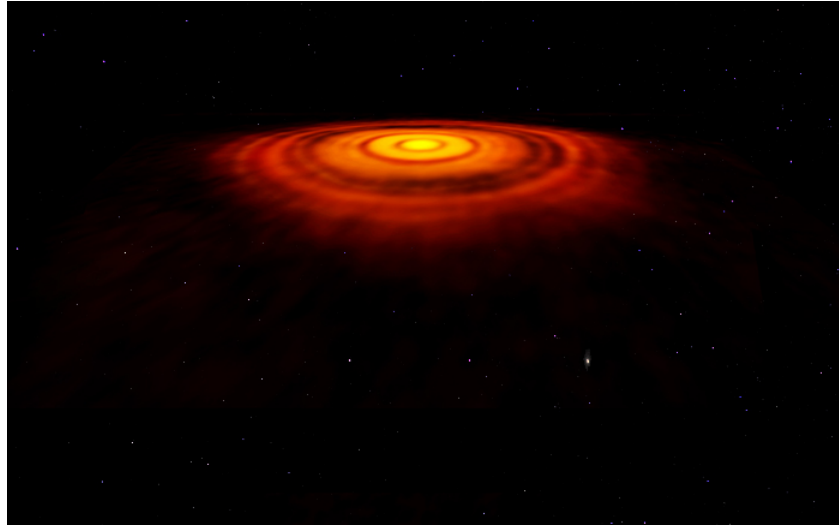


Figure 2.1.3: An artist's conception of an oblique view of a protoplanetary disk, using actual photoimagery of the [HL Tauri](#) disk, originally gathered by the European Southern Observatory.

Once a star forms in the center, astronomers call the ring of debris around it a **protoplanetary disk**. For our solar system, this happened 4.55 billion years ago.

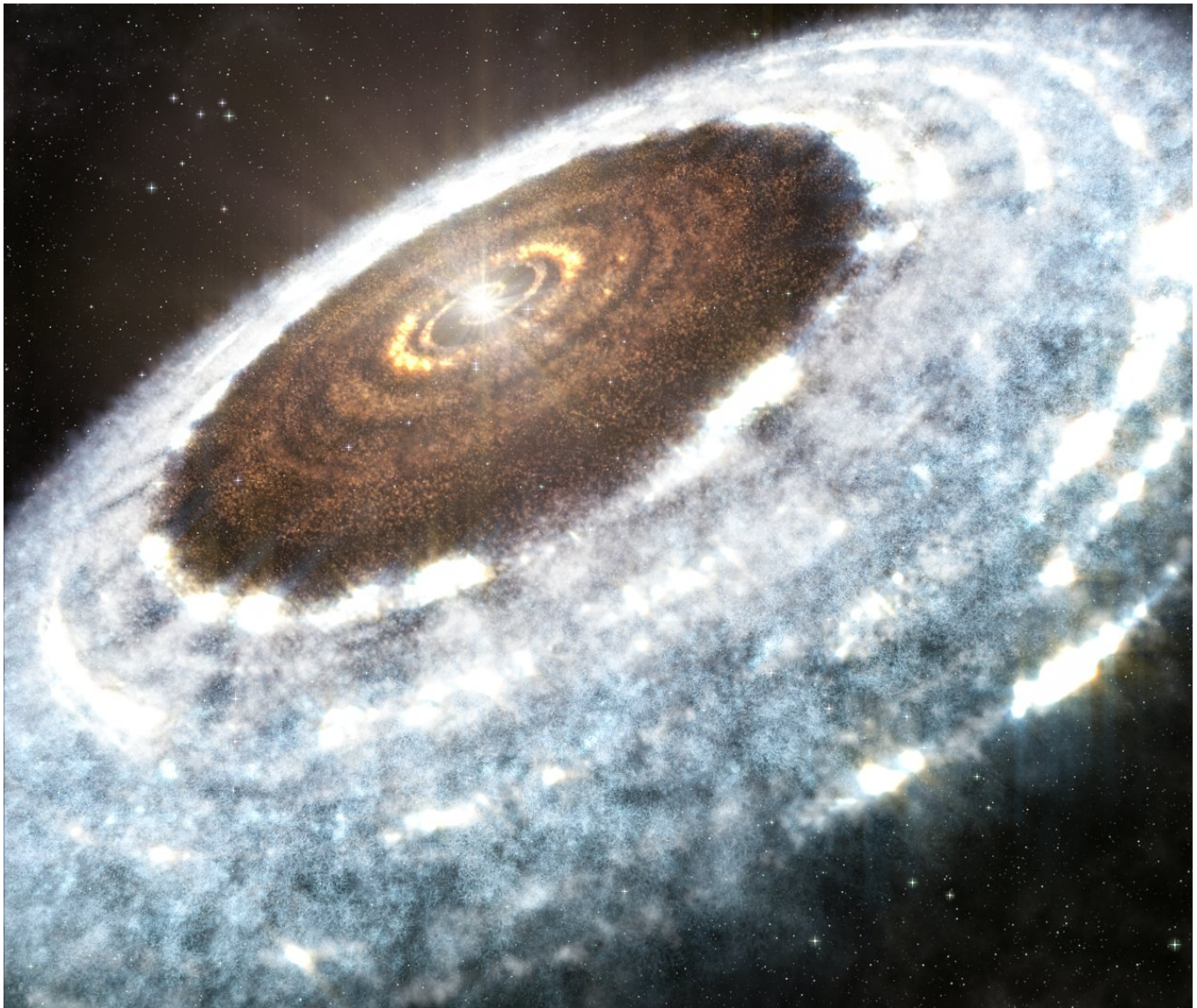


Figure 2.1.4: This artist's impression of the water snowline around the young star V883 Orionis, as detected with ALMA.

Two important processes that helped further organize the protoplanetary disk and form planets were condensation and accretion.

Condensation is a process that allows gaseous matter to stick together to make liquid or solid matter. We have evidence of condensation in the form of small spherical objects with internal layering, kind of like “space hailstones.” These are **chondrules**, and they represent the earliest objects formed in our solar system. (Occasionally, we are lucky enough to find chondrules that have survived until today, entombed inside certain meteorites.)

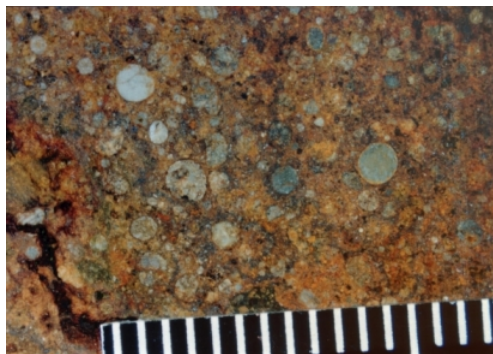


Figure 2.1.5: Chondrules in a meteorite, with a scale in mm.

Chondrules glommed onto other chondrules and stuck themselves together into primordial “rocks,” building up larger and larger objects. Eventually, these objects got to be big enough to have sufficient gravity to pull their mass into a round shape, forming **planetesimals**, the nuclei of planets. Planetesimals gobbled up nearby asteroids and smashed into other planetesimals, merging and growing through time through **accretion**. The kinetic force of these collisions heated the rocky and metallic material of the planetesimals, and their temperature also went up as radioactive decay heated them from within. Once warm, **differentiation** could happen, where denser material sank to their centers (or cores), and lighter-weight elements and compounds rose to their surface (or outer layers).

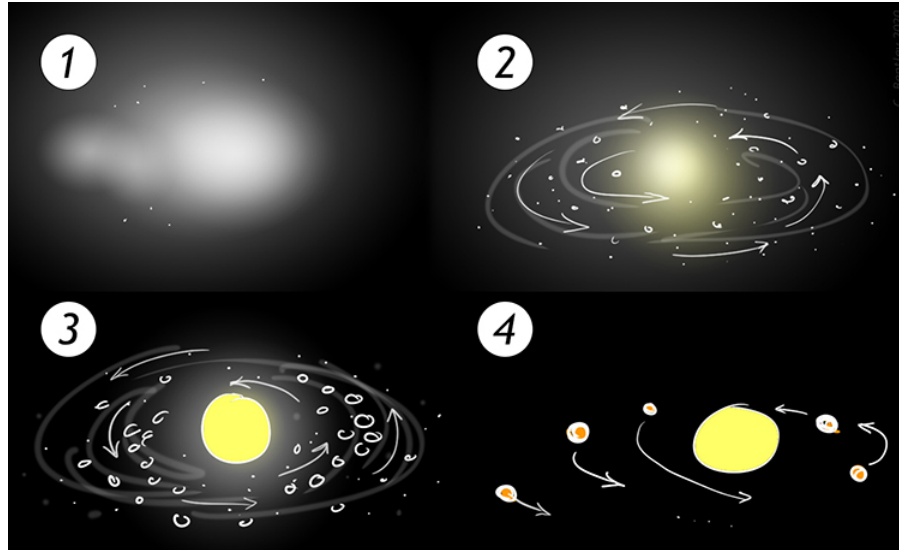


Figure 2.1.6: A cartoon model showing the evolution of our solar system from a pre-solar nebula.

Metallic iron meteorites, such as the Tamentit meteorite below, are used as evidence of the differentiation of planetesimals into layered bodies. Meteorites with metallic compositions represent the “core” material from these planetesimals.



Figure 2.1.7: Tamentit Iron Meteorite, found in

1864 in the Sahara, weighing about 500 kg (1,100 lb). From Ji-Elle, Public domain, via Wikimedia Commons.

The process of accretion continues into the present day, though at a slower pace than the earliest days of the solar system. You can observe this in the asteroid belt, where certain asteroids are simply a clump of space rocks, held together by their gravity. Consider the asteroid called Itokawa 25143, for instance which is made up of numerous uniquely-sized boulders rather than of a single solid body.



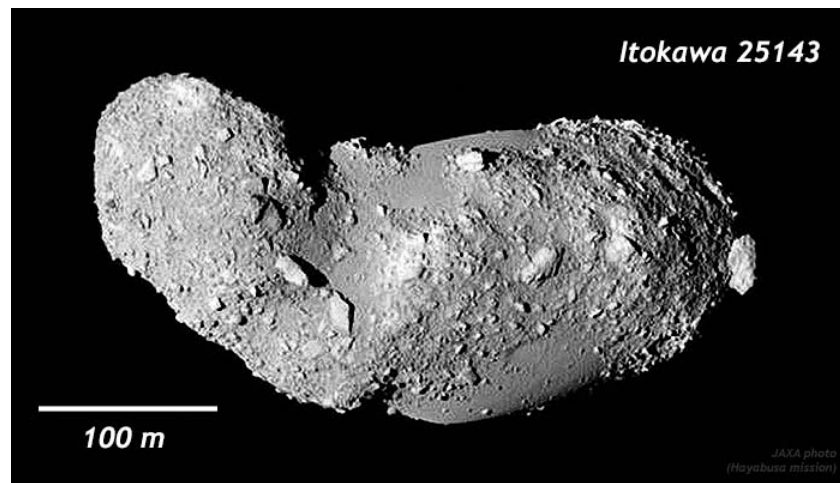


Figure 2.1.8: The asteroid 25143 Itokawa.

Another modern example of accretion is meteorite impacts. Every time a chunk of rock in space intersects the Earth, its mass is added to that of the planet. In that instant, the solar system gets a little bit cleaner (fewer leftover bits rattling around) and the planet gets a little more massive. A spectacular example of this occurred in 1994 with Comet Shoemaker-Levy 9 when Jupiter's immense gravity broke the comet into chunks, then swallowed them up one after another. Astronomers on Earth watched with fascination as the comet chunks, some more than a kilometer across, slammed into Jupiter's atmosphere at 60 km/second (~134,000 mph), creating a 23,700 °C fireball and enormous impact scars that were as large as the entire Earth. These scars lasted for months.

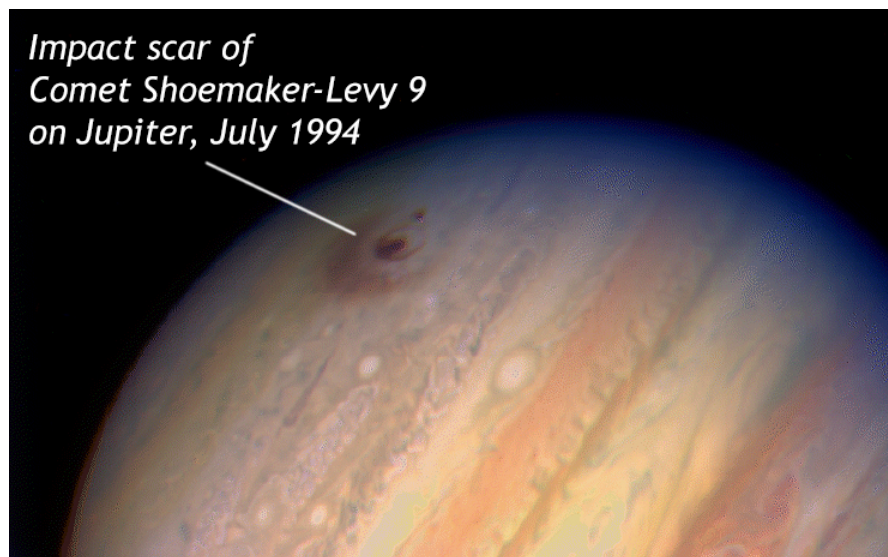


Figure 2.1.9: Accretion in action: scar from the July 1994 collision between a fragment of Comet Shoemaker-Levy 9 and the surface of the atmosphere of the planet Jupiter.

This incredibly dramatic event reminded Earthlings that they are not safe from accretionary impacts, just as the dinosaurs discovered some 66 million years ago. In this case, an object from the solar system was removed while adding a bit to Jupiter's mass.

In summary, Earth is part of a solar system centered on the Sun. This solar system, with its star, planets, dwarf planets, and “leftover” comets and asteroids, formed from a nebula full of elements in the form of gas and dust. Over time, these very small pieces stuck together to make larger concentrations of mass, eventually culminating in a star and its orbiting planets. Asteroids (and asteroids that fall to Earth, called meteorites) are leftovers from this process. The original nebula formed from the destruction of some previous star that exploded in a supernova - a galactic-sized example of recycling!

 Key Terms

- **accretion** - increasing the mass of an object by adding material to it
- **chondrules** - small grains found in certain types of meteorites formed in the early solar nebula as molten droplets quickly cooled and crystallized
- **differentiation** - material settling out according to density
- **nebula** - a massive cloud of gas and dust in space that is the byproduct of a previous star that suffered a supernova explosion
- **planetesimals** - the nuclei of planets formed in the early solar system from collisions with other objects in the solar system
- **protoplanetary disk** - ring of debris around a star

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## 2.2: Earth's Structure and the Origin of the Continents and Oceans

### Forming Earth's Structure

Around 4.55 billion years ago, the brand new Earth was extraordinarily hot and mostly molten, resembling a ball of lava floating in space.

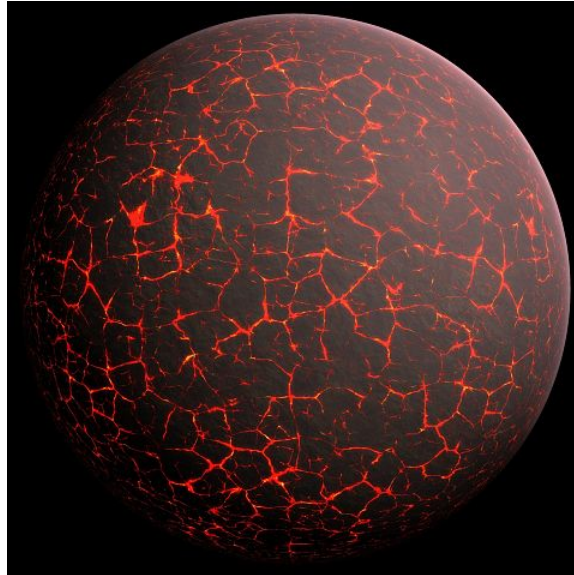


Figure 2.2.1: Earliest Earth.

As Earth cooled from its molten state, minerals crystallized and settled according to their densities - differentiation transformed a homogenous sphere of magma into a heterogeneous planet with a distinct structure. Metals and radioactive elements have the greatest density and sank to the center to make the core, while silicate compounds floated to the surface. Iron and magnesium-rich silica compounds of moderate density took up the space between. This resulted in three primary layers: the core, mantle, and crust. The terms **lithosphere** and **asthenosphere** are also used to describe the structure of Earth's outer layers. The lithosphere includes the solid uppermost mantle and crust, while the asthenosphere describes the semi-solid upper mantle below. These terms are particularly important in the context of **plate tectonics** which is discussed in detail elsewhere in this book. For now, consider it the theory that Earth's rigid outer shell, the lithosphere, is broken up into individual pieces called plates, that move and interact with one another along plate boundaries.

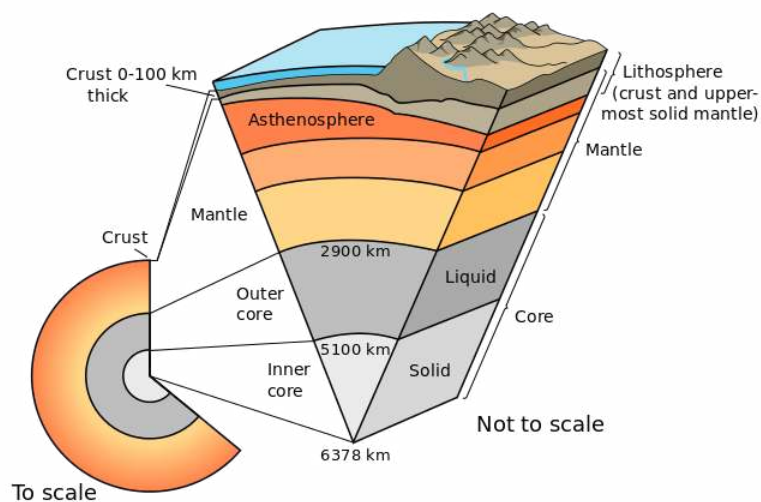


Figure 2.2.2: The structure of Earth. The crust, mantle, and core are layers with distinct compositions. Lithosphere and asthenosphere have different physical properties.

## Origin of the Continents

The first continental crust formed shortly after the earth's origin, but continents, as we know them today, and the plate tectonic processes that build and shape them did not happen until much later. Initially, crustal pieces were continually split apart by heat rising from Earth's very hot core. Earth's surface may have resembled the lava lakes we can see today within the crater of a few volcanoes.

### Lava Lake Video

Watch the video from 1:31 to 2:03.



## Origin of Earth's Water

The origin of surface water on Earth is still uncertain. Explanations for its origin include volcanic outgassing, comets, and meteorites. The volcanic outgassing hypothesis for the origin of Earth's water is that it originated from inside the planet, and emerged via tectonic processes as vapor associated with volcanic eruptions. Since all volcanic eruptions contain some water vapor, at times more than 1% of the volume, these alone could have created Earth's surface water. Another likely source of water was from space. Comets are a mixture of dust and ice, with some or most of that ice being frozen water. Seemingly dry meteors can contain small but measurable amounts of water, usually trapped in their mineral structures.

### Where Did Our Water Come From? Video

Watch the video from 4:47 to 10:03.



During heavy bombardment periods later in Earth's history, its cooled surface was pummeled by comets and meteorites, which could be why so much water exists above ground. There isn't a definitive answer for what process is the source of ocean water. Earth's water isotopically matches water found in meteorites much better than that of comets. However, it is hard to know if Earth processes could have changed the water's isotopic signature over the last 4-plus billion years. It is probably that all three sources contributed to the origin of Earth's water.



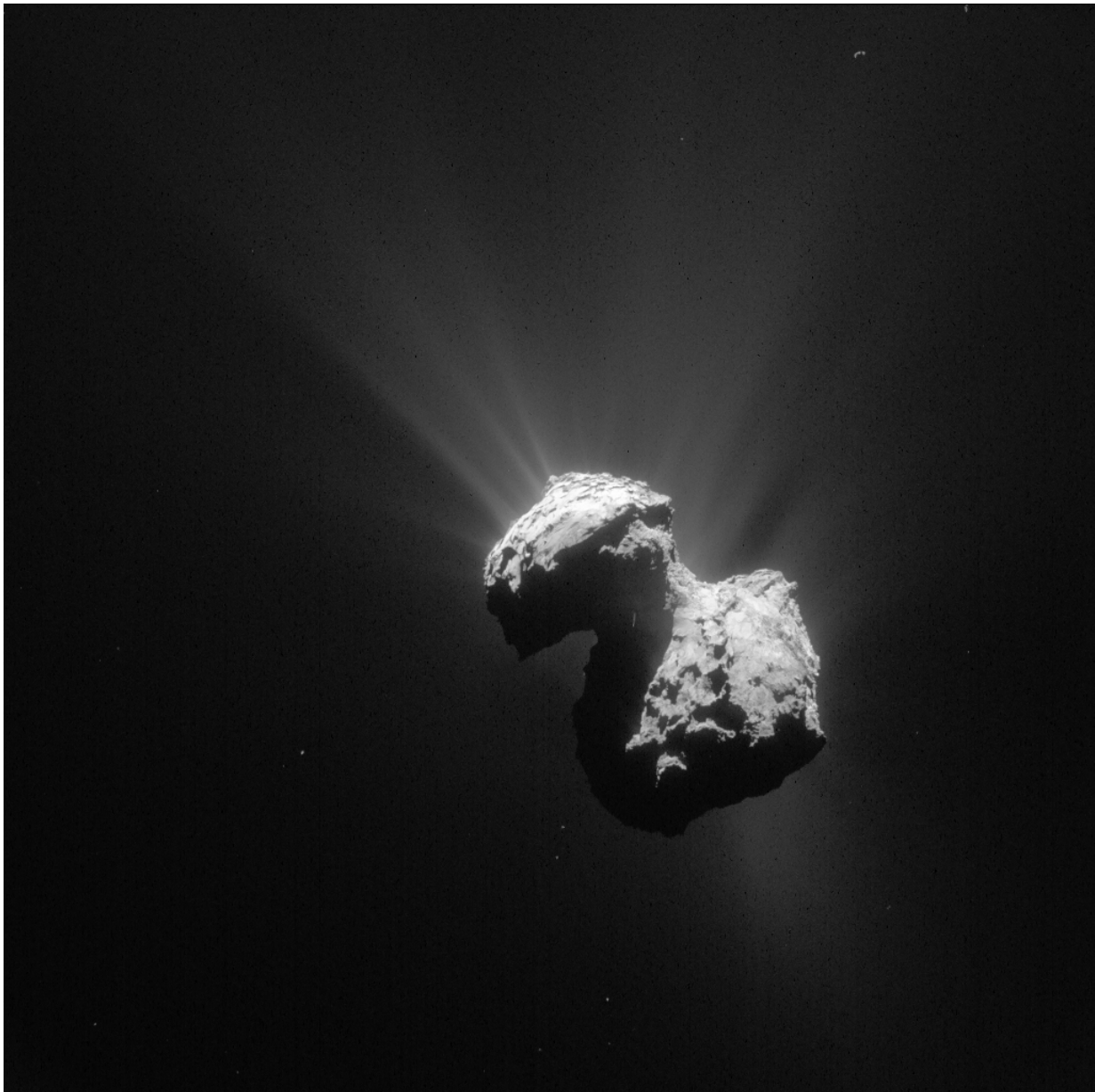


Figure 2.2.3: Water vapor leaves comet 67P/Churyumov-Gerasimenko.



#### Key Terms

- **asthenosphere** - the layer beneath the lithosphere, which is in a plastic, semi-solid state
- **lithosphere** - the outermost layer of Earth that includes the crust and solid upper mantle; this solid, rock-layer, is broken up into individual pieces called plates
- **plate tectonics** - the theory that Earth's rigid outer shell, the lithosphere, is broken up into individual pieces called plates, that move and interact with one another along plate boundaries, driving Earth's constructive geologic processes

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## 2.3: Key Points

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- Earth and our solar system formed 4.55 billion years ago from a nebulae
  - The Earth's layered structure is the result of differentiation, the separation of materials according to density
  - Earth's crust formed shortly after the origin of the planet but continents as we know them didn't happen until much later
  - Oceans developed from the outgassing of volcanoes and meteor and comet impacts
- 

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## CHAPTER OVERVIEW

### 3: Plate Tectonics

#### Introduction

In the late 1960s and early 1970s, geoscientific consensus gelled around an intriguing idea: that many different observations about our planet could be explained with a single model of Earth dynamics. The idea of “plate tectonics” put together old ideas about continental drift (continents move around on top of the ocean floor) with new data showing seafloor spreading (seafloor is created at "cracks" in the ocean floor as the two sides separate). The new theory was a revolutionary idea that made the distribution of volcanoes, earthquakes, continents, and oceans make sense in a way that no idea had achieved before. Fifty years of additional data and testing have confirmed plate tectonics as the key idea in modern geology.

[3.1: Plates Defined](#)

[3.2: Tectonics Defined](#)

[3.3: Plate Boundaries and Interiors](#)

[3.4: Geophysical Phenomena](#)

[3.5: Driving Forces](#)

[3.6: The Beginning of Plate Tectonics and the Continents](#)

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[3.8: Key Points](#)

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